J Mater Sci: Mater Electron **(2025) 36:115**

Exploring the morphological and optical properties of N-doped ZnO heterojunctions

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Received: 11 October 2024
Accepted: 22 December 2024
Published online: 9 January 2025

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ABSTRACT

Nitrogen-doped zinc oxide (N:ZnO) thin films were deposited on glass substrates via radio frequency (RF) magnetron sputtering and subsequently annealed at 300 °C, 400 °C, 500 °C, and 600 °C to assess their viability and stability as transparent conductive oxide (TCO) materials. Structural and compositional analyses were performed using X-ray diffraction (XRD), field emission scanning electron microscopy (FESEM), and X-ray photoelectron spectroscopy (XPS). XRD analysis revealed preferential crystallite orientations along the (100), (002), (101), and (110) planes. Atomic force microscopy (AFM) measurements indicated particle sizes two to four times larger than those derived from XRD, suggesting a subgranular internal structure, as XRD probes coherently diffracting domains. XPS analysis of the N 1 s spectra identified two distinct peaks at approximately 397 eV and 407.5 eV, indicating nitrogen incorporation into the ZnO matrix. Photoluminescence spectroscopy revealed that nitrogen doping induced the formation of interstitials and defects associated with oxygen and zinc vacancies. Optical measurements showed that the (N:ZnO) thin films exhibited an average optical band gap of approximately 3.1 eV, with 80% transmittance in the visible spectrum. A linear relationship was observed between the band gap energy and the tail width. Except for the film annealed at 600 °C, all annealed films showed a reduction in peak photoluminescence intensity with increasing annealing temperature. Finally, no significant changes in the electrical performance of the p-N/n-Si

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https://doi.org/10.1007/s10854-024-14150-7

diode were observed as a result of annealing-induced surface modifications. The results provide valuable insights into the optimization of (N:ZnO) thin films for use in international optoelectronic and photovoltaic research, where advancements in TCOs are critical for the development of high-performance, sustainable technologies.

1 Introduction

Zinc oxide (ZnO) is a wide-bandgap semiconductor (Eg = 3.37 eV) with a high free exciton binding energy (60 meV), making it highly attractive in material science, physics, chemistry, biochemistry, and solid-state electronics. Its notable properties—such as optical transparency, bandgap tunability, and the ability to undergo heavy n-type doping—make ZnO ideal for a wide range of optoelectronic and structural applications [1, 2].

Quantum mechanical size effects, combined with a considerable increase of the surface to bulk ratio, are responsible for the unique properties of nanometer-sized particles, including electronic and geometric structure, and optical and magnetic properties [3]. A thorough understanding of the dependence of these properties on particle size is of great importance notably for the design of next generation materials [4, 5].

ZnO nanostructures have garnered significant research attention due to their promising potential in photonic devices, displaying high optical gain (320 cm^{-1}) at room temperature. As a result, ZnO is considered a candidate for transparent electrode materials in optoelectronic devices, sensors, and spintronics [6, 7]. Furthermore, ZnO has emerged as a promising material for ultraviolet light-emitting diodes (LEDs) and lasers [8, 9].

ZnO-based p-n heterojunctions, including core–shell structures, nanorods, and thin films, have been extensively studied, with thin film heterojunctions playing a critical role in photovoltaic and optoelectronic applications [8]. However, the performance of devices is limited by challenges in achieving reliable p-type ZnO films. Enhancing the crystalline quality while reducing background carrier concentration is crucial for optimizing ZnO homojunction luminescence. Post-growth annealing is a known method to improve film properties, including electrical behavior, luminescence efficiency, and crystal quality [10, 11]. Despite this, the high activation energy and low solubility of acceptor dopants in ZnO hinder p-type conduction.

Group-V elements (N, P, As) and group-I elements (Li, Na) have been explored as potential acceptor dopants. Among them, nitrogen is the most favorable candidate for substituting oxygen in ZnO due to its similar atomic radius (1.68 Å for nitrogen versus 1.38 Å for oxygen) and comparable valence 2p states, as well as its lower activation energy compared to other Group-V elements [10, 12]. Nitrogen substitution introduces shallow acceptor states (NO) with a hole binding energy of 200 meV. Studies indicate that lower oxygen pressures during sputtering favor NO formation, as they reduce the formation energy of NO defects. Studies also indicate that nitrogen oxides (NO, NO_2) are more efficient sources of nitrogen for doping than N_2O or N_2, as they lower the energy barrier for N–N bond dissociation [10, 13].

Nitrogen-doped ZnO thin films with varying dopant concentrations have been synthesized using various methods, including wet chemistry. These films have been analyzed using photoluminescence (PL), Raman spectroscopy, optical reflection, and X-ray diffraction (XRD), with heavily doped ZnO films revealing a novel phase. It means that when the concentration of N dopant in ZnO film increases, the new peaks marked by asterisk signal appear, which implies the detection of a new phase [14]. Minor shifts in Raman peaks indicated stress within the ZnO layers. RF magnetron sputtering has also been employed to deposit nitrogen-doped layers on quartz substrates, using a mixture of N_2 and O_2 gases at 773 K. After annealing, the N:ZnO films exhibited p-type conduction, as confirmed by Hall effect measurements. Temperature-dependent photoluminescence (PL) spectra revealed enhanced near-band-edge (NBE) emission, further indicating the presence of an acceptor energy level [15]. Previous studies have shown that the microstructure, composition, and functional properties of thin films are significantly affected by the deposition technique [16–20]. RF magnetron sputtering offers several advantages over other deposition methods, including simplicity, cost-effectiveness, and the ability to operate at lower processing temperatures. However, achieving high-quality film deposition requires careful

consideration of factors such as substrate properties, sputtering atmosphere, deposition parameters, and post-processing conditions [21, 22].

The issue of instability of the ZnO doped with N and its P-type is very important and has been discussed in previous articles, and improving the quality of these thin films is also discussed [23].

In this work, we prepared good quality and stable thin films that have n-p diode behavior by post-annealing in a nitrogen gas environment immediately after growth. Because post-annealing is an effective and suitable process for improving crystal structures and reducing the concentration of background carriers or electrical behaviors and luminescence properties.

In this study, we employed RF magnetron sputtering to deposit p-type N:ZnO heterojunction thin films on quartz substrates, incorporating nitrogen as a dopant during the deposition process. The films were then annealed in a nitrogen gas flow at four distinct temperatures to investigate their structural and optoelectronic properties. This article is organized as follows: after the introduction the experimental detail is described in Sect. 2. Section 3 is devoted to the results and discussion, which includes three subsections such as structural properties, optical properties and electrical properties of N:ZnO thin films and the conclusion is shown in Sect. 4.

2 Experimental details

In this study, quartz substrates measuring 1×2 cm^2 with a thickness of 4 mm were utilized. Initially, each sample was cleaned with soap and water to eliminate surface contaminants. Following this, the substrates were subjected to ultrasonic cleaning in acetone and ethanol baths for 10 min each. To avoid any further contamination, the samples were individually stored in clean containers. The deposition of nitrogen-doped ZnO (N:ZnO) thin films was performed using an RF magnetron sputtering system (MSS160) with a ZnO target. Sputtering was conducted at a power of 110 W for 60 min, under a working pressure of 6.5×10^{-3} mbar and a base pressure of 1.4×10^{-4} mbar. A gas mixture of 30% nitrogen (N_2) and 70% argon (Ar) was used as the sputtering atmosphere. The resulting thin films had a thickness of 300 nm. To assess the durability and stability of the N:ZnO thin films, they were annealed at four different temperatures: 300 °C, 400 °C, 500 °C, and 600 °C. The annealing process was conducted in an N_2 gas flow for one hour, after which the samples were gradually cooled to room temperature. A summary of the sample details is provided in Table 1.

The X-ray diffraction (XRD) patterns were obtained using a Philips PW3710 diffractometer with Cu-Kα radiation (wavelength $\lambda = 0.154$ nm). XRD data were analyzed to determine the crystalline structure and phase composition of the N:ZnO thin films deposited on various substrates and annealed at different temperatures. Phase identification was carried out using X'Pert software. To investigate the average grain size, distribution, and changes induced by annealing, atomic force microscopy (AFM) was performed using a Park Scientific Instruments CP AutoProbe in contact mode. Grain and particle size distributions were further analyzed with WSxM software. The reflectance and transmittance spectra of the annealed samples were recorded using a Varian Cary 500 Scan UV–Vis NIR Spectrophotometer. Photoluminescence (PL) spectroscopy was employed to evaluate the luminescent properties of the films, with measurements performed using a Cary Eclipse fluorescence spectrometer. The excitation source was a xenon lamp, and the excitation wavelength for the N:ZnO samples was set at 320 nm. Additionally, the surface morphology and structural features of the films were captured using a Field Emission Scanning Electron Microscope (FESEM), model MIRA3 TESCAN-XMU. On the other hand,

Table 1 Optimal growth conditions and annealing parameters of N:ZnO thin films on quartz substrates

Annealing temperature (°C)	Base pressure (mbar)	Work pressure (mbar)	Sputtering rate (A°/s)	Sputtering gas	Thickness (nm)
300 400 500 600	1.4×10^{-4}	6.5×10^{-3}	0.83	30% N_2–70 % Ar	300 ± 5

the FPP-5000 Miller Inc., Four Point Probe resistivity meter was used to measure the surface electrical resistivity of films.

3 Results and discussion

3.1 Structural properties of N:ZnO thin films

The semi-quantitative elemental analysis of the N:ZnO nanostructures was performed using Energy Dispersive X-ray Spectroscopy (EDX). The EDX spectrum revealed two prominent peaks corresponding to the L shell of zinc (Zn) and the K shell of oxygen (O), along with a weaker peak corresponding to the K shell of nitrogen (N), confirming the successful incorporation of nitrogen into the ZnO matrix and the formation relative composition N:ZnO nanostructures. These results are consistent with previously reported findings in Reference [19]. The EDX spectra of N:ZnO films annealed at temperatures of 300, 400, 500, and 600 °C are shown in Fig. 1. The accompanying bar diagram in Fig. 1d displays the atomic percentages of Zn, O, and N across the different annealing temperatures. While the atomic percentage of oxygen remained constant, slight variations were observed in the atomic percentages of zinc and nitrogen. Notably, the percentage of nitrogen increased with higher annealing temperatures. However, this increase in nitrogen content did not significantly impact the electrical and optical properties of the N:ZnO thin films, as will be further discussed.

The crystalline structure of N:ZnO thin films was characterized using X-ray diffraction (XRD). The resulting diffraction spectra (Fig. 2a) were analyzed to derive the grain size distribution (GSD), extending the classical Scherrer equation to address polygranular materials [25]. This extended method enables the estimation of the average grain size D_{av} and its

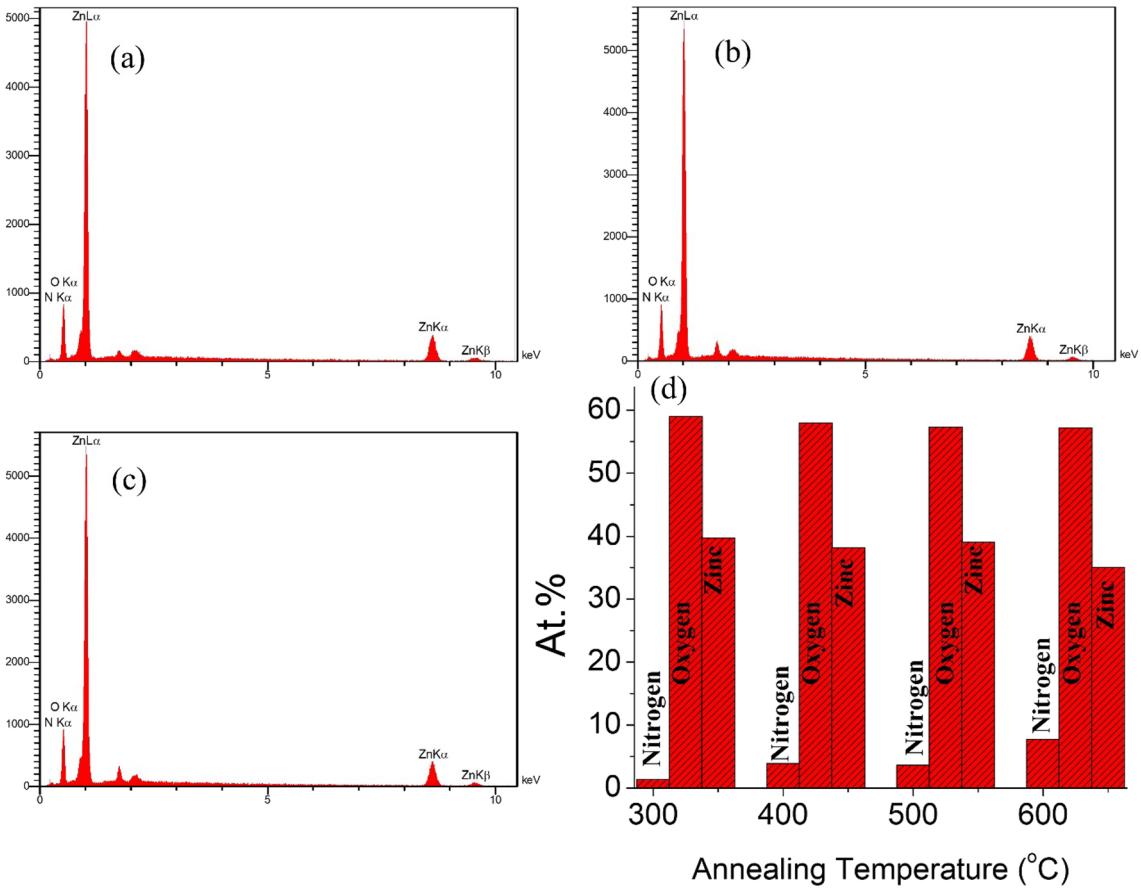


Fig. 1 **a** EDX spectrum of N:ZnO annealed at 300 °C, **b** EDX spectrum of N:ZnO annealed at 500 °C, **c** EDX spectrum of N:ZnO annealed at 600 °C, and **d** Atomic percentages of Zn, O, and N as determined from the EDX spectra

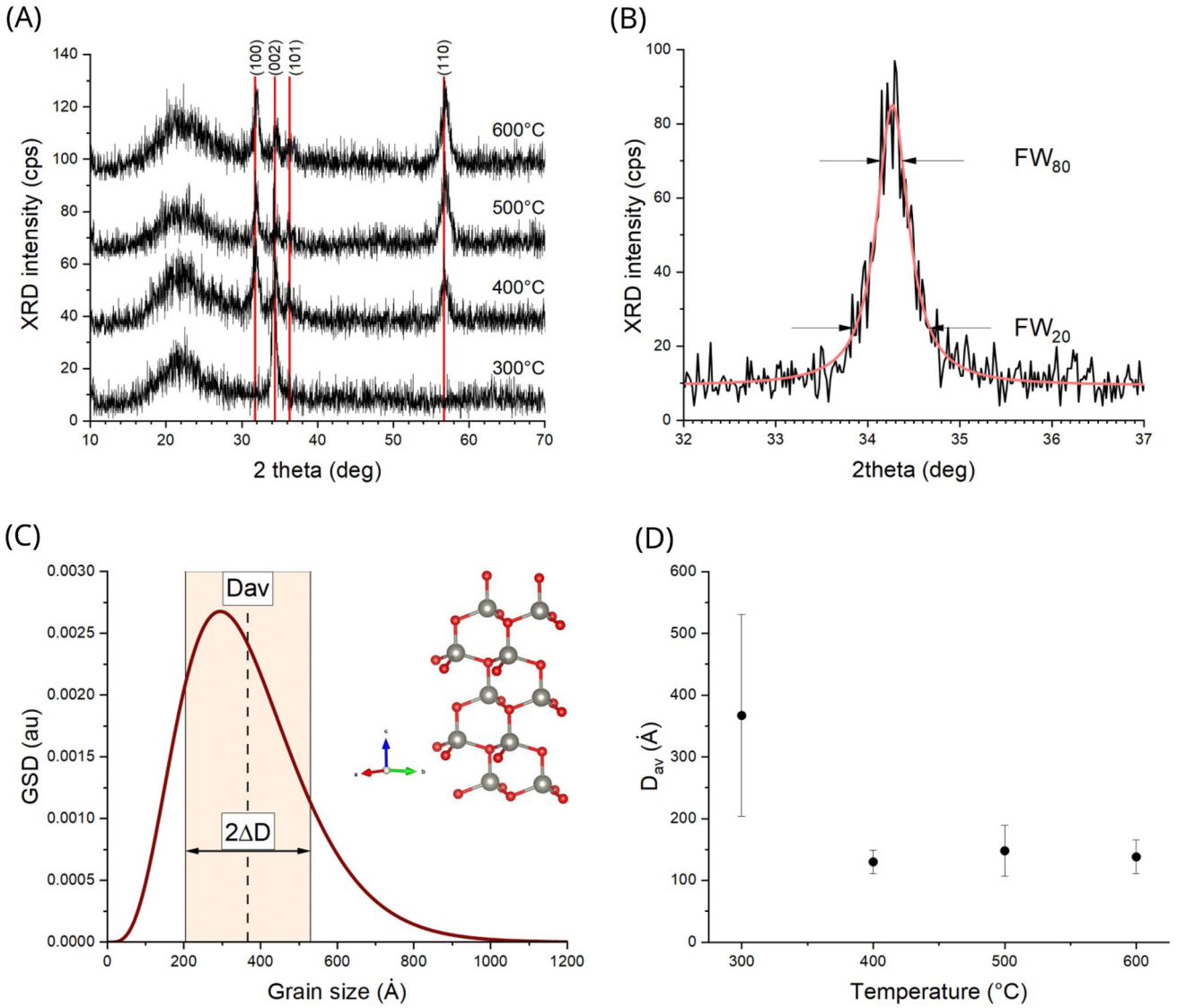


Fig. 2 **A** XRD spectra of N:ZnO thin films deposited at various temperatures. **B** XRD (002) peak for the film deposited at 300 °C (black) fitted with a Pearson VII profile (red). **C** Plot of the grain size distribution (GSD) profile, showing an average grain size D_{av} = 36.7 nm and dispersion ΔD = 16.3 nm, corresponding to the (002) peak of the film deposited at 300 °C. Inset: ZnO crystal lattice visualized using VESTA [22]. **D** Scatter plot illustrating the average grain size D_{av} (including size dispersion ΔD) as a function of deposition temperature

dispersion ΔD, based on the assumption of a grain size distribution function [26]:

$$GSD(x, \rho, \mu) = \frac{x^{\mu} \exp(-x/\rho)}{\rho^{\mu+1}\Gamma(\mu+1)} \quad (1)$$

The parameters of interest are associated with the auxiliary variables ϱ and μ as follows [26]:

$$D_{av} = \rho(\mu+1) \quad (2)$$

$$\Delta D = \rho\sqrt{\mu+1}, \quad (3)$$

where: x represents the grain diameter, and Γ is the Euler gamma function.

The grain size D_{av} and its dispersion ΔD were determined using the linewidths of a single XRD peak, measured at different heights. To simplify the analysis, the full widths at 20% and 80% of the peak height (denoted as FW_{20} and FW_{80}) were used. The XRD data for a single peak were fitted with a Pearson

VII function to obtain these linewidths (Fig. 2b). The derived GSD curve for the (002) peak of the N:ZnO film is shown in Fig. 2c, displaying both the mean grain size D_{av} and size dispersion ΔD. This approach allows a more nuanced understanding of grain size distribution in polycrystalline thin films.

To validate the accuracy of the grain size distribution (GSD) approach, the results were compared with those obtained from the Scherrer equation [27]:

$$D_S = \frac{0.9\lambda}{\beta \cos(\theta)}, \quad (4)$$

where λ is the X-ray wavelength, β represents the line broadening due to finite domain size, and θ is Bragg's angle.

Additionally, the size of the diffracting domains within the lattice was estimated using the Williamson-Hall method, denoted as D_{WH}. The obtained data are summarized in Table 2.

At 300 °C, the average particle size (D_{av}) is 36.698 nm, which is the highest among the measured temperatures. As the temperature increases to 400 °C, D_{av} significantly decreases to approximately 12.906 nm, indicating that higher annealing temperatures result in smaller particle sizes. The trend continues with D_{av} values around 14.810 nm and 14.063 nm for 500 °C and 600 °C, respectively. This suggests that initial growth may lead to larger grains, which then become finer at higher temperatures due to processes such as thermal diffusion and recrystallization. The size dispersion (ΔD) values indicate the uniformity of the particle sizes. At 300 °C, ΔD is 16.285 nm, reflecting significant variation among the grain sizes. This value decreases at 400 °C and remains relatively low at higher temperatures, suggesting improved uniformity of the grain sizes with increasing temperature. The D_S values are generally smaller than the D_{av} values from GSD, which is expected as the Scherrer equation typically provides a lower estimation of grain size due to its dependence on peak broadening. For instance, at 300 °C, D_S is 25.582 nm, while D_{av} is 36.698 nm. This difference highlights the impact of grain size distribution and crystallographic contributions to peak broadening in the XRD patterns. The D_{WH} values are also consistently lower than the D_{av} values, indicating that both the Williamson-Hall method and the Scherrer equation are affected by factors such as lattice strain in addition to particle size. The values for D_{WH} range from 10.507 nm to 11.756 nm across the temperature variations, indicating that strain plays a significant role in the observed diffraction patterns. Overall, the trends indicate that higher temperatures may facilitate particle coarsening and enhance uniformity, but they also reveal that the interplay of size and strain significantly influences the diffracting domains of N:ZnO thin films.

The surface topography of the N:ZnO films was analyzed using atomic force microscopy (AFM) images to estimate particle size. The Otsu's thresholding algorithm was employed to classify the data values

Table 2 Diffracting domain size established from XRD spectra of N:ZnO films by means of three different methods

Temperature (°C)	hkl	2θ (deg)	D_{av} (nm)	ΔD (nm)	D_S (nm)	D_{WH} (nm)
300	002	34.33	36.698	16.285	25.582	-
400	100	31.82	12.906	1.650	11.316	11.756
	002	34.46	12.962	1.679	11.279	
	101	36.31	13.811	2.439	11.255	
	110	56.69	12.445	1.619	11.012	
500	100	31.83	14.810	3.973	10.843	11.329
	002	34.42	14.554	2.972	10.801	
	101	36.31	15.455	5.101	10.771	
	110	56.71	14.321	4.087	10.507	
600	100	31.81	14.063	2.833	11.082	11.538
	002	34.43	13.379	1.889	11.044	
	101	36.30	14.280	3.090	11.018	
	110	56.68	13.530	2.880	10.767	

D_{av} – average particle size from GSD, ΔD – size dispersion from GSD, D_s – average particle size due to Scherrer, D_{WH} – average particle size due to Williamson-Hall

into two distinct classes by minimizing the intra-class variances. The resulting images are displayed in Fig. 3, with the estimated average grain diameters for the samples deposited at various temperatures being 69.2 nm at 300 °C, 53.4 nm at 40 °C, 59.9 nm at 500 °C, and 52.1 nm at 600 °C.

In comparison to the sizes obtained from X-ray diffraction (XRD), the particle sizes determined from atomic force microscopy (AFM) measurements are found to be larger by a factor ranging from 2 to 4. This discrepancy may reflect the sub-granular inner structure of the particles, as AFM captures surface features while XRD scattering occurs within coherently diffracting domains, which are typically more continuous in nature. Figure 4 shows field emission scanning electron microscopy (FESEM) images that were analyzed for surface morphology using statistical shape descriptors. The analysis began with binarizing the images using the watershed algorithm, followed by the determination of several parameters that describe the planar figures: circularity, roundness, solidity, and Feret's diameter.

Circularity is defined as a positive fractional value that quantifies the deviation of a shape from a perfect circle. It is calculated using the following formula [8]:

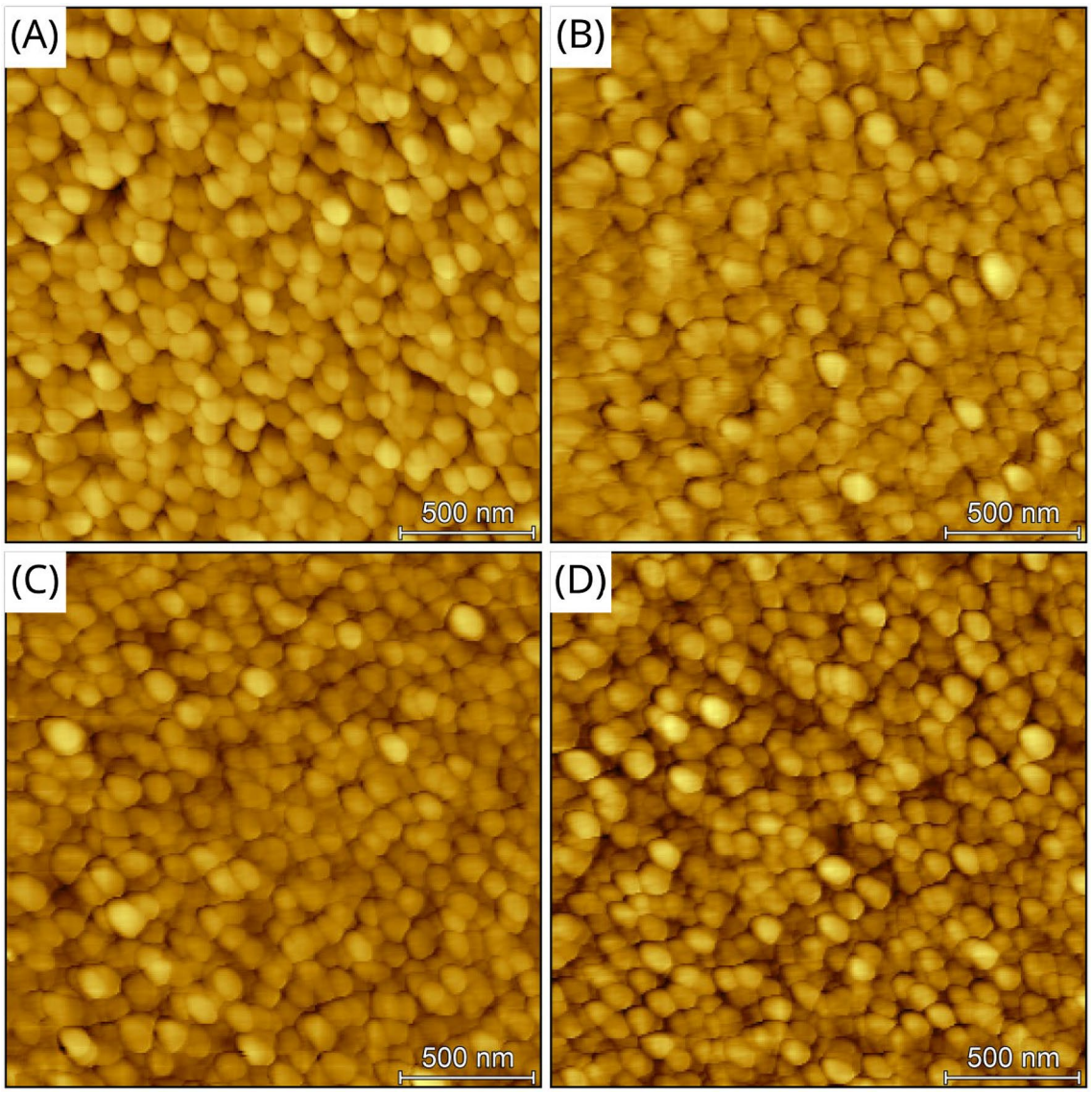


Fig. 3 AFM images of surface topography of N:ZnO films deposited at various temperatures: **A** 300 °C, **B** 400 °C, **C** 500 °C, **D** 600 °C

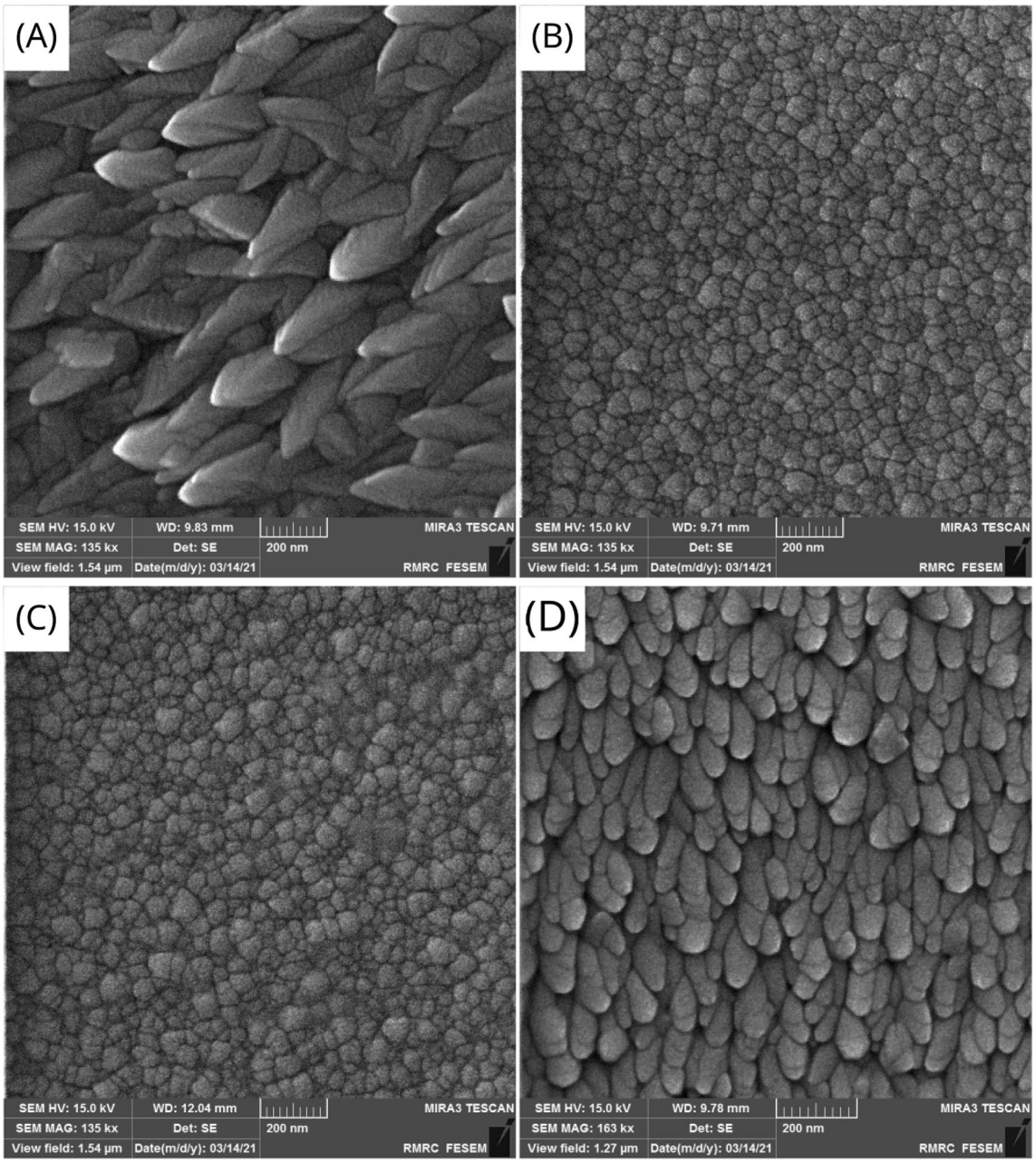


Fig. 4 FESEM images of surface topography of N:ZnO films deposited at various temperatures: **A** 300 °C, **B** 400 °C, **C** 500 °C, **D** 600 °C

$$C = \frac{4\pi A}{P^2}, \tag{5}$$

where the segment area is denoted by A and its perimeter by P.

As the circularity approaches zero, the shape becomes increasingly elongated; conversely, as it approaches unity, the shape resembles a perfect circle. Roundness, defined as the ratio of the lengths of the major and minor semi-axes of the best-fit ellipse that approximates the selected area, serves as a comparable metric [9].

$$R = \frac{a_{\min}}{a_{\max}} = \frac{A}{\pi a_{\max}^2}, \tag{6}$$

where: a_{min}, a_{max} are the minor and major semi-axes of the equivalent ellipse, respectively. In turn, solidity is

the ratio of the actual area of the figure and its convex hull [9]:

$$S = \frac{A}{A_{CH}} \tag{7}$$

For a perfectly convex shape, solidity is equal to one; for non-convex shapes, it remains less than one but greater than zero. This parameter provides specific measurements of the segments in terms of various lengths. The Feret diameter d_F is defined as the length of a straight line connecting any two points on the boundary of the segment. Among the various d_F values, the minimum and maximum Feret diameters are particularly significant for characterizing the shape and form of the particles.

Figure 5 shows half-box plots illustrating the parameters derived from the segmentation data obtained from FESEM images. The average Feret diameters were measured at 220 nm, 61 nm, 65 nm, and 110 nm for samples deposited at 300 °C, 400 °C, 500 °C, and 600 °C, respectively. Notably, the data for the sample deposited at 300 °C exhibit a greater spread compared to the other samples, indicating significant inhomogeneity in crystal sizes (Fig. 5a). This variability suggests that the crystal growth at this temperature may be less controlled, leading to a wider distribution of particle dimensions. Additionally, the Feret diameter for the 300 °C sample is approximately three times larger than the corresponding estimate derived from the AFM image, while the factor for the 600 °C sample is closer to two. This discrepancy may stem from the tilted surface of the samples, which can result in an elongation of the columnar crystals as seen in Figs. 4a, d. Such elongation affects the perceived size when measured at an angle, leading to inflated diameter estimates. In contrast, for the samples deposited at 400 °C and 500 °C, the results from both measurement methods align closely, indicating a more consistent and uniform crystal morphology. Furthermore, when illuminated along the surface normal, this effect is minimized, providing clearer cross-sectional views of the growing crystals, which allows for more accurate estimations of their diameters. This highlights the importance of illumination direction and surface orientation in accurately assessing particle size and morphology using imaging techniques.

The shape descriptors presented in Fig. 5b circularity, roundness, and solidity—indicate that the particles exhibit relatively regular shapes. The average circularity and roundness values, which range between 0.6 and 0.7, suggest that the particles have an almost circular morphology. In contrast, the solidity values, which fluctuate around 0.9, indicate that the particles are nearly perfectly convex. Collectively, these findings demonstrate that the particles possess a consistent shape, akin to those observed in the AFM images.

In this study, we investigated the effects of nitrogen incorporation into the ZnO lattice, focusing on

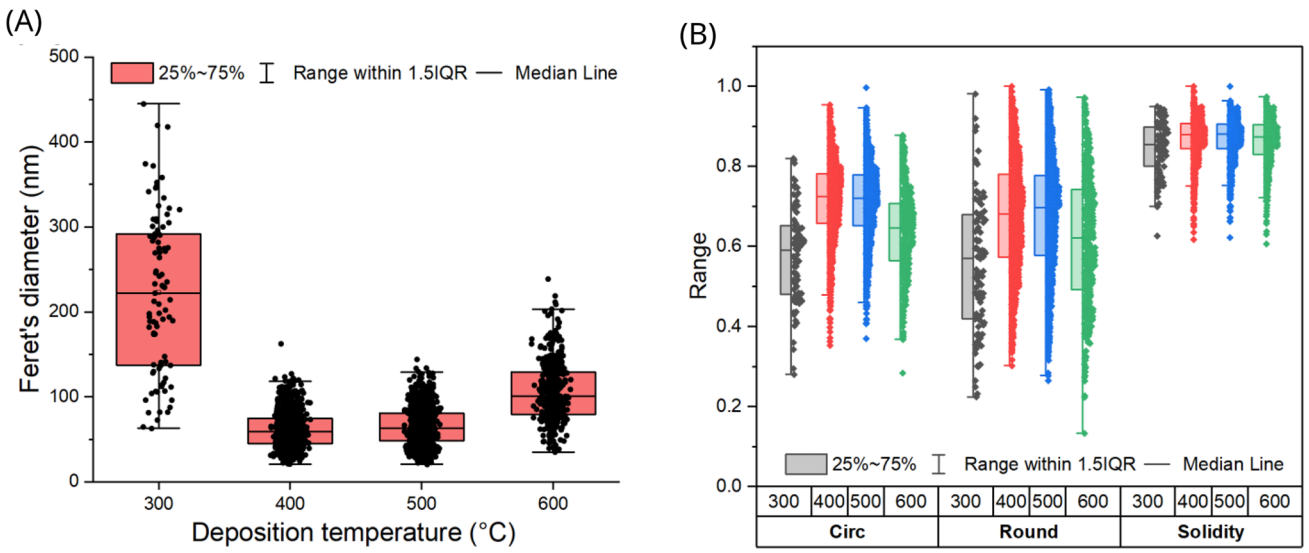


Fig. 5 The parameters derived from segmentation data from FESEM images **A** Feret's average diameters at various temperatures. **B** shape descriptors

the chemical bonding and electronic structure of the resulting films. This was accomplished by analyzing the X-ray photoelectron spectroscopy (XPS) core-level and valence-band spectra. Notably, the XPS spectra for all samples at various annealing temperatures exhibited minimal variation, suggesting a uniform chemical environment across the films.

As illustrated in Fig. 6, the O 1 s spectrum reveals a prominent peak at approximately 530.5 eV. Additionally, the survey XPS spectra for each film display lines corresponding to Zn 2p3/2 and Zn 2p1/2 at binding energies of 1031 eV and 1054 eV, respectively (Fig. 6b). These binding energies align with those of the ZnO wurtzite structure's matrix elements, which possess similar XPS core levels [22].

Weak N 1 s signals within the energy range of 395–410 eV are also evident in the XPS spectra of the N:ZnO films (Fig. 6c). The N 1 s spectra reveal two components, likely arising from different chemical bonding environments of the nitrogen atoms. The low-energy component of the N 1 s core-level spectrum, observed at approximately 397 eV, corresponds to the formation of chemical N-Zn bonds resulting from the substitution of nitrogen atoms for oxygen atoms in the N:ZnO film. Conversely, the high-energy component, detected at approximately 407.5 eV, is attributed to N–O bonds, which may result from nitrogen atoms adsorbed onto the oxide surface [28, 29].

The presence of both N–Zn and N–O bonds in the N:ZnO thin films confirms the incorporation of nitrogen into the ZnO structure. Furthermore, the high atomic percentage of oxygen, as determined from the EDX analysis, supports the formation of these bonds, indicating a complex interplay between nitrogen incorporation and oxygen content during the annealing process.

3.2 Optical properties of N:ZnO thin films

The characteristics of the radiation emission spectra of nitrogen-doped zinc oxide (N:ZnO) were investigated alongside their photoluminescence spectrum, utilizing an excitation wavelength of 320 nm provided by a xenon lamp. This analysis aimed to elucidate the optical properties and emission behavior of the N:ZnO films under specified excitation conditions.

The emission spectra of nitrogen-doped zinc oxide (N:ZnO) thin films revealed four distinct peaks corresponding to ultraviolet (UV), violet, blue, and green

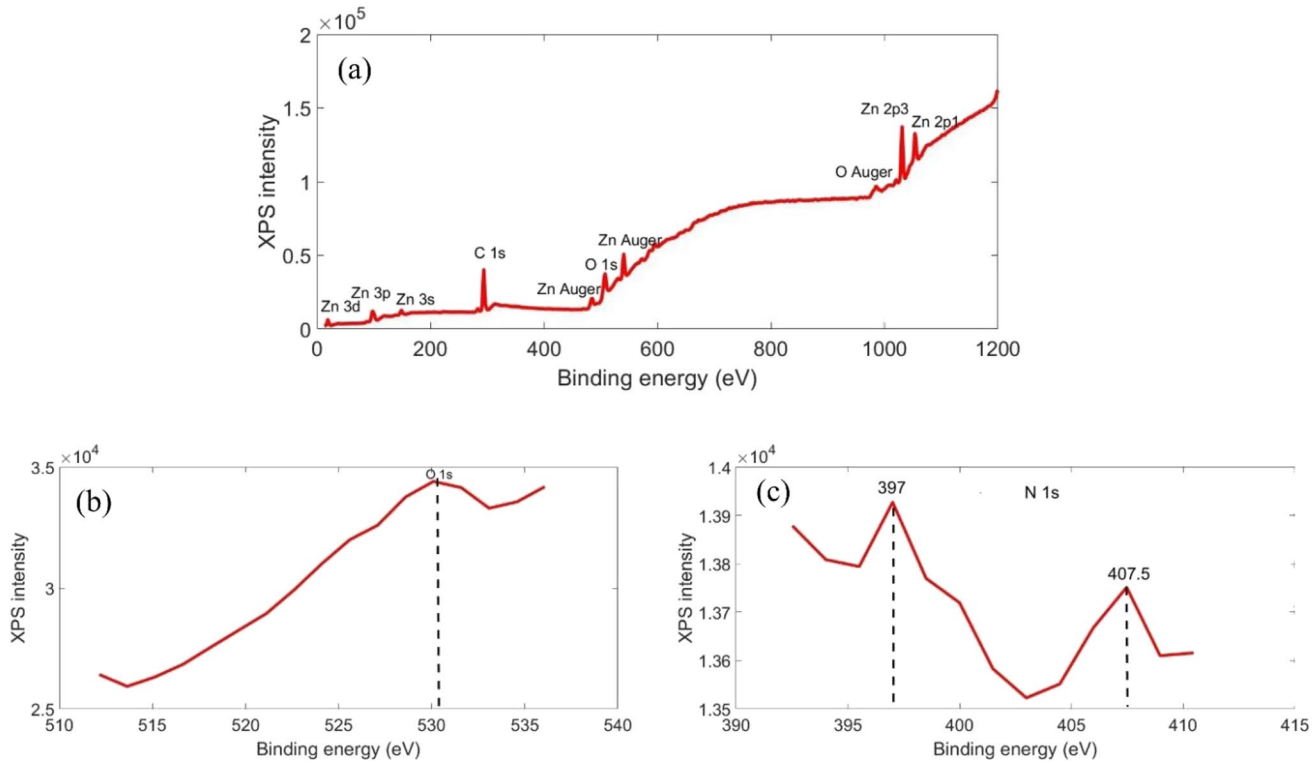


Fig. 6 XPS survey spectrum of the N:ZnO annealed at 500 °C: **a** full spectrum, **b** high-resolution curve of O 1 s, **c** high-resolution curve of N 1 s

emissions, as illustrated in Fig. 7. The UV emission, observed between 395 and 400 nm, is attributed to the Near Band Edge (NBE) transitions, while the other peaks fall within the Deep Level Emission (DLE) category. Specifically, the UV emission results from the recombination of free excitons in the ZnO lattice. A reduced concentration of inherent holes within the ZnO layer limits exciton formation, thereby diminishing the intensity of the UV peak.

When nitrogen ions substitute for oxygen ions in the ZnO crystal structure, they act as acceptors, increasing the availability of holes and excitons. Excitons, which are bound pairs of electrons and holes, are significantly influenced by Coulomb interactions. The quantum confinement effect in N enhances exciton formation at the interface by raising the exciton binding energy. This modification of electron and hole wave functions facilitates efficient exciton generation, even in the presence of a shortened mean free path typically associated with charge carriers in the material.

The intensity of the photoluminescence (PL) spectrum serves as a relative indicator of defect density within the N:ZnO thin films. The violet emission displays two peaks around 420 nm and 402 nm, attributed to zinc vacancy defects (VZn) associated with deeper acceptor states of VZn, which are considered dominant in contributing to the p-type conductivity of N:ZnO thin films. Additionally, a prominent peak located at 487 nm is identified as deep-level emission, likely related to oxygen (O)-deficient defects [14]. As the annealing temperature increases, the intensity of the emission spectra also increases, indicating a rise in defect density. However, for samples annealed at 300 °C and 600 °C, the PL intensity decreases similarly, suggesting these two samples exhibit comparable crystal defect characteristics and lower density of states, resulting in fewer defects overall.

The intensity of the photoluminescence (PL) spectrum increases as a relative indicator of defect density within the N:ZnO thin films with increasing annealing temperature. Specially, oxygen (O)-deficient defects that related a peak located at 487 nm, when nitrogen ions substitute for oxygen ions in the ZnO crystal structure, they act as acceptors, increasing the availability of holes and excitons.

To further explore the optical properties of the N:ZnO thin films, transmission and reflection spectra were analyzed to derive various optical parameters. The transmission spectrum, depicted in Fig. 8, demonstrates the transparent nature of the N:ZnO thin films within the visible region, achieving transparency levels of approximately 90% with increasing annealing temperature. Moreover, the absorption edge of the thin films exhibits a slight redshift towards longer wavelengths as the annealing temperature rises, indicating changes in the electronic structure and possible modifications in the optical bandgap of the films.

The reflection spectrum of the N:ZnO layers, presented in Fig. 9, exhibits fluctuations and remains below 35%. This oscillatory behavior is attributed to interference effects occurring between the two interfaces: the boundary between the N:ZnO layer and air, as well as the interface between the layer and the underlying substrate. These interfacial interactions contribute to the observed variability in the reflection spectrum.

The absorption coefficients of the thin films are calculated by the relation $\alpha = \frac{1}{d} Ln \frac{1}{T}$, where T is the optical transmittance; d is the thickness and α is the absorption coefficient.

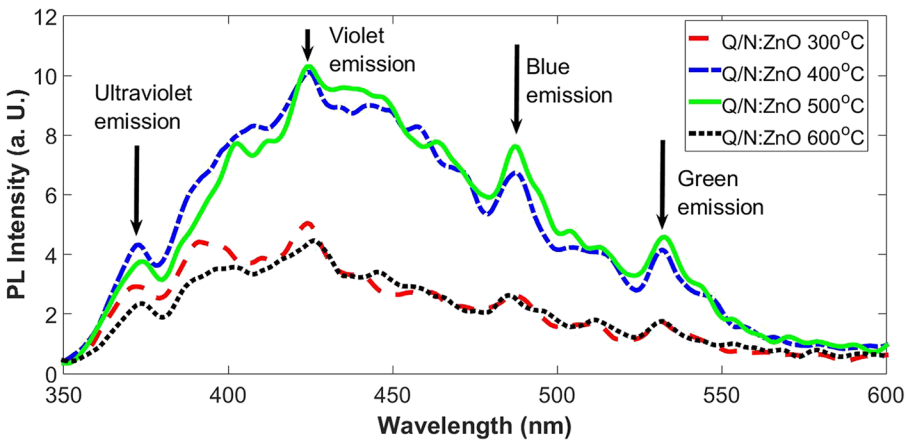


Fig. 7 PL spectrum of N:ZnO thin films annealed at different temperatures

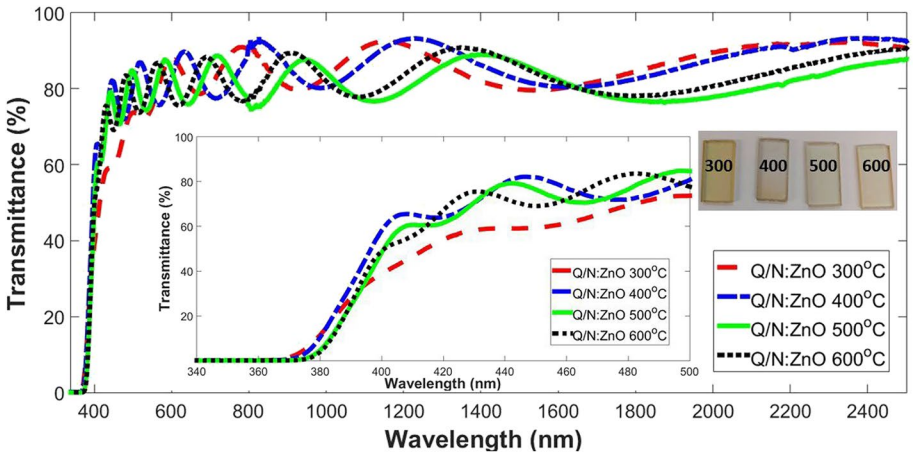


Fig. 8 Transmission spectrum of N:ZnO thin films on quartz substrate with annealing at different temperatures

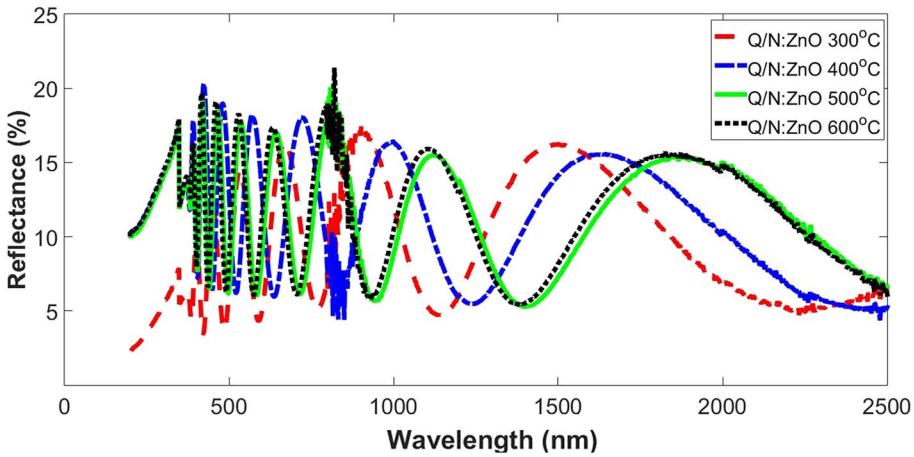


Fig. 9 Reflection spectrum of N:ZnO thin films on quartz substrate with annealing at different temperatures

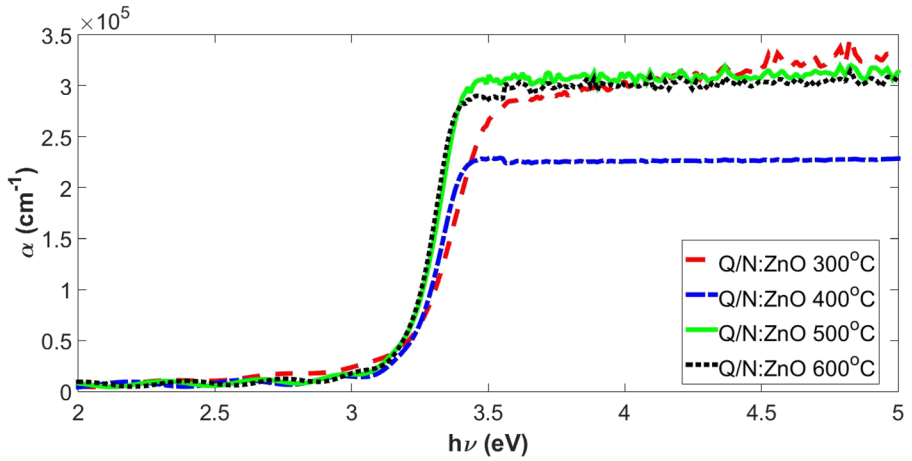


Fig. 10 Absorption coefficient of N:ZnO thin films annealed at different temperatures

Figure 10 illustrates the variations in the absorption coefficient as a function of the incident light energy. The changes in the absorption coefficient with increasing annealing temperature exhibit a slight red shift. These alterations are hypothesized to be influenced by the size of the N:ZnO nanoparticles.

The optical band gap of N:ZnO thin films deposited on quartz substrates is calculated using the following Eq. (8) [21]:

$$(\alpha h\nu)^{1/m} = C(h\nu - E_g), \tag{8}$$

where α is the absorption coefficient, hν is the absorbed photon energy, and E_g is the optical gap energy, and C is a constant independent of photon energy. The exponent m indicates the type of electronic transition. The plot of $(\alpha h\nu)^{1/m}$ versus photon energy is known as the Tauc diagram, which facilitates the determination of the optical band gap value by extending a fitted line to the point where $(\alpha h\nu)^{1/m} = 0$. The various values of m correspond to different transition types: $m = 2$ for direct transitions, $m = 1/2$ for indirect transitions, $m = 3$ for non-direct transitions, and $m = 2/3$ for indirect non-allowed transitions.

Additionally, the Urbach energy (E_u), which represents the width of the tail of localized states in the band gap, can be calculated by plotting Ln(α) against the photon energy. The relationship is described by the equation [21]:

$$\alpha = \alpha_0 \exp(\frac{h\nu}{E_u}), \tag{9}$$

where E_u is the Urbach energy and α_0 is a constant that depends on temperature. The sequence of the widths of substituted states or the width of the absorption edge can be used to determine E_u.

Figure 11 displays the photon energy diagram as a function of $(\alpha h\nu)^2$ for the samples, from which the band gap values are derived and presented in Table 3. Meanwhile, Fig. 12 illustrates the photon energy plot used for calculating Urbach energy. The Urbach energy quantify the static, structural disorder causing localized exponential-tail states, and dynamic disorder from electron–phonon scattering. According to inset Fig. 12., the value of ΔE is the electronic width near the edges of the conduction band and the valence-band, which depends on the Urbach energy. The value of ΔE in the band gap can be calculated by the Urbach energy. The Urbach energy (Eu) is the width of the localize states and is defined in semiconductors where the absorption in the band gap is broadened due to various defects in the crystal structure. The absorption edge of N:ZnO thin films becomes narrower with increasing annealing temperature. The enhancement in the localize states in the band gap leads to larger amounts of energy due to non-radiative recombination

Table 3 Optical band gap results of N:ZnO thin films on quartz substrate

Sample	E_g(eV)	E_u (meV)	E_g(e mission) (eV)
N:ZnO—300 °C	3.15	150.34	3.33
N:ZnO—400 °C	3.15	112.42	3.32
N:ZnO—500 °C	3.14	109.03	3.31
N:ZnO—600 °C	3.14	96.64	3.31

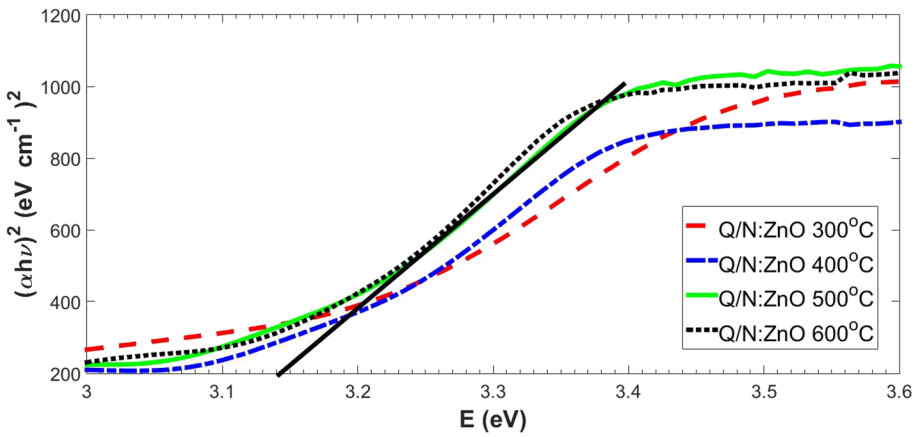


Fig. 11 Determination of the optical band gap of N:ZnO thin films annealed at different temperatures

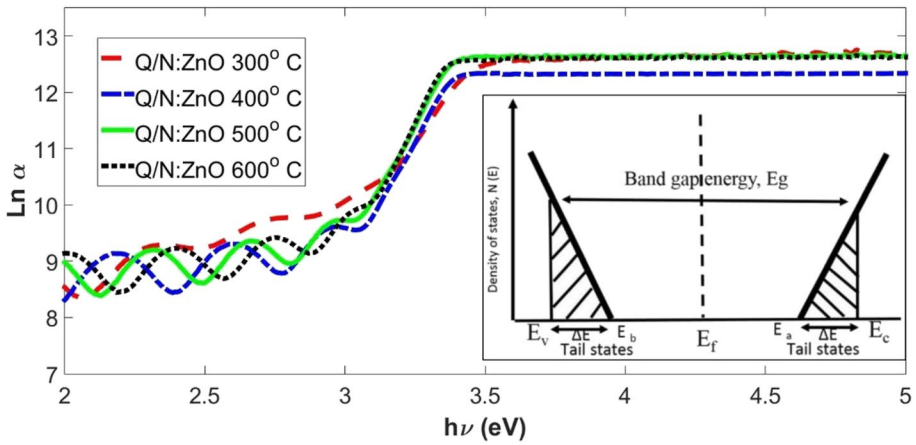


Fig. 12 *Lnα* of N:ZnO thin films annealed at different temperatures and inset: Schematic diagram of the density of states and energy

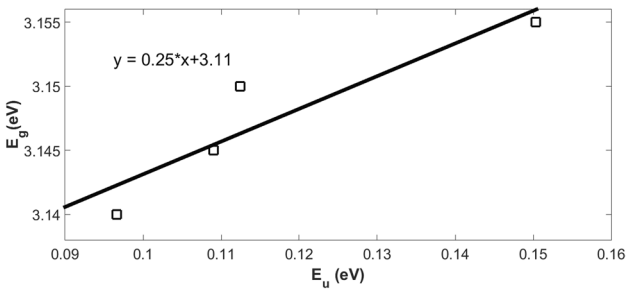


Fig. 13 The relationship between the band gap energy and the width of the Urbach energy

centers. As expected, vacancies and dislocation defects in the layers are responsible for introducing the localize states into the band gap; which can drastically change the optical absorption properties of the layer. The curvature of the absorption edge encompasses many different optical properties of thin film materials, including localize states and disorder effects, and amorphous and polycrystalline materials have a high concentration of localize states, grain boundaries, and various defects that impair crystallinity. The optical parameters, including the band gap energy E_g and Urbach energy E_u, are summarized in Table 3, showing that E_u decreases with increasing annealing temperature. To establish a clearer linear relationship between E_u and E_g, these parameters are plotted together in Fig. 13, revealing a linear correlation that can be expressed by the following linear adaptation equation [10]:

$$E_g = 0.25E_u + 3.11 \quad (10)$$

The optical band gap energy E_g for the N:ZnO films shows a slight decrease with increasing annealing temperature, ranging from 3.15 eV at 300 °C to 3.14 eV at 600 °C. This consistent reduction in E_g could be attributed to enhanced structural and electronic interactions as the nitrogen atoms incorporate into the zinc oxide lattice. The substitution of nitrogen for oxygen in the ZnO structure may result in localized states near the conduction band, influencing the band gap narrowing. The emission energy values show minimal variation across samples, with values slightly higher than the corresponding optical band gaps. This indicates that the optical transitions involved in the photoluminescence process are influenced by other factors such as defects and excitonic interactions. The values are consistently around 3.31 eV to 3.33 eV, suggesting that despite the changes in structural properties with annealing, the intrinsic emission characteristics of the material remain stable. The Urbach energy, which is an indicator of the degree of disorder and the width of the tail of localized states in the band gap, decreases significantly from 150.34 meV at 300 °C to 96.64 meV at 600 °C. This reduction suggests an improvement in crystallinity and a decrease in defect density with higher annealing temperatures. A lower Urbach energy correlates with narrower distribution of localized states, indicating that the films are becoming more ordered as the annealing temperature increases.

The optical values indicate that variations in annealing temperature have only slight effects on the structural and optical parameters of the nitrogen-doped zinc oxide (N:ZnO) thin films. This observation suggests that the N:ZnO films demonstrate

stability when subjected to temperature changes ranging from 300 °C to 600 °C. Given the similarities in optical and structural characteristics across the different annealing temperatures, we selected the sample annealed at 500 °C on a silicon substrate for subsequent electrical testing.

3.3 Electrical properties of N:ZnO thin films

Surface four-point probe analysis was used to confirm the p-type of N:ZnO thin films and their surface resistivity values.The surface electrical resistivity of all samples showed P-type and their values ranged from 0.6×10^4 to 0.9×10^4.

Figure 14 shows the current as a function of voltage for the N:ZnO-p /Si–n structure. The current–voltage (I–V) characteristics reveal that the N:ZnO thin film behaves similarly to a typical diode structure. In this figure, the I–V characteristics for the p–n junction of the N:ZnO thin film deposited on the silicon substrate at 500 °C are displayed under both dark and illuminated conditions. The schematic of the p–n junction formed between the n-Si and p-ZnO layers is also included. In this configuration, electron diffusion occurs from the n-Si region to the p-ZnO region, while hole diffusion takes place from the p-ZnO to the n-Si region. A space charge region at the p–n junction plays a critical role in limiting the current flow at low forward bias voltages by maintaining a depletion zone that impedes charge carrier movement. As the forward bias increases, the depletion region narrows, resulting in an exponential increase in current. The electric field strength within the junction significantly influences the device's performance by affecting charge carrier mobility and recombination rates.

Figure 14 further illustrates the relationship between brightness and current–voltage characteristics, highlighting how illumination impacts the I-V readings of the device. This relationship underscores the importance of understanding the optical and electrical interactions in N:ZnO films for optimizing their performance in optoelectronic applications.

The n value of N:ZnO thin films was calculated using Cheung's method [24] with the following relations:

$$\frac{dV}{d(LnI)} = \frac{nKT}{q} + IR_s \tag{11}$$

$$H(I) = V - \frac{nKT}{q} Ln\left(\frac{I}{AA^*T^2}\right) \tag{12}$$

$$H(I) = n\varphi_b + IR_s \tag{13}$$

Using this technique, an ideal n-factor of 1.23 was determined for the N:ZnO diode sample that was annealed at 500 °C. The deviation of the n-factor from the ideal value of 1 suggests the presence of a surface layer between the metal and semiconductor, as well as additional current conduction mechanisms that may include tunneling, recombination, and other processes beyond simple thermal diffusion. This behavior is indicative of a heterojunction, where different semiconductor materials are joined, creating potential barriers that affect charge transport. For each sample, the effective barrier height of the Schottky junction (ϕb) was measured to be approximately 0.3 eV. This barrier height is a critical parameter that influences the device's electrical characteristics, particularly its rectification properties. Diode ideality factors(n-factor) are

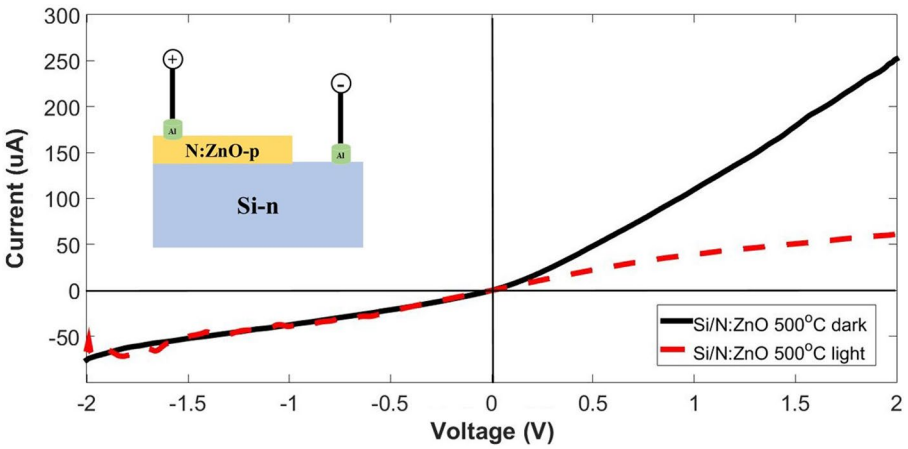


Fig. 14 I-V diagram of N:ZnO thin film for silicon substrate at 500 °C in darkness and light

Table 4 The diode parameters of N:ZnO-p/Si–n annealed at 500 °C heterojunction structures and the related literature data

Sample	$I_{SC}(\mu A)$	Plot dv/d(lnI) vs I		Plot H(I) vs I	
		n	R_s (Ω)	φ_b (eV)	R_s (Ω)
Our work	0.514	1.23	0.822×10^4	0.373	1.046×10^4
Keskenler et al. [31]	3.62	1	–	0.81	198
Akay et al. [32]	0.036	4.95	92	0.63	76/88
Bedia et al. [33]	0.007	2.5	–	0.90	6.28×10^3

expected to have the values of 1.0–2.0 whose archives values are noticeably higher. There are some reasons that have been ascribed to the contribution of the Schottky barrier of metal–semiconductor interfaces or trap-assisted tunneling, as well as carrier leakage [14]. For instance, the high ideality factor could be attributed to trap-assisted tunneling or quantum-well and carrier leakage due to the defects in the N:ZnO/Si interface.

Table 4 summarizes the series resistance values, which were calculated from the slope of the curve (dV/dln(I). The analysis of the p–n junction characteristics unequivocally demonstrates that the annealed N:ZnO thin films exhibit p-type conductivity. This conclusion is supported by the observed rectification behavior, with a threshold voltage of 0.50 V, confirming the expected performance associated with p-type semiconductors. Such rectification is a fundamental property of p-n junctions, enabling them to effectively control the flow of electrical current, which is essential for various electronic applications.

4 Conclusion

In this study, nitrogen-doped zinc oxide N:ZnO thin films were successfully fabricated on quartz substrates using radio frequency (RF) reactive sputtering and subsequently annealed at temperatures of 300, 400, 500, and 600 °C. The films were analyzed and classified based on their electrical and structural properties, tailored for specific applications. The N:ZnO thin films exhibited crystallization in the wurtzite phase, characterized by the presence of the (100), (002), (101), and (110) planes. Various methods, including grain size determination (GSD), Scherrer analysis, and Williamson-Hall techniques, were employed to estimate the grain size from X-ray diffraction (XRD) spectra, revealing the size of the diffracting domains, which represent the smallest structural units with continuity. XRD analysis revealed when the peak intensity of preferred direction (002) is more than peaks of (100) and (101) the nitrogen concentrations will be less in N:ZnO thin films. Therefore, at the annealing temperature of 400 °C, when the (100) and (101) crystal planes appeared, it indicates a higher concentration of nitrogen in the thin films and the placement of more nitrogen in the ZnO structure because the presence of the (100) and (101) crystal planes indicate oxygen vacancy defects and interstitial Zn atoms, which increases the probability of the presence of nitrogen. Additionally, atomic force microscopy (AFM) was utilized to investigate the grain size and boundaries, while field emission scanning electron microscopy (FESEM) provided insights into the non-homogeneous nature of the material. Interestingly, the largest grain size was observed in the sample deposited at 300 °C, while higher annealing temperatures resulted in a reduction of grain size by factors ranging from 0.5 to 0.8, depending on the estimation method, indicating a temperature-independent trend. The binding energies of the intense O 1 s lines (≈ 530.5 eV) were consistent with those of the matrix elements of the ZnO wurtzite structure, corroborating the findings from X-ray photoelectron spectroscopy (XPS). The optical band gap of the N:ZnO thin films was found to be approximately 3.1 eV for all samples, with a transmittance of about 80% in the visible wavelength range. A linear relationship was identified between the tail width and the band gap energy. Notably, the peak photoluminescence intensity exhibited a decreasing trend with increasing annealing temperatures, except for the sample annealed at 600 °C, which displayed an increase in spectral intensity, suggesting enhanced defect density. The intensity of the photoluminescence (PL) spectrum increases as a relative indicator of defect density within the N:ZnO thin films with increasing annealing temperature. Specially, oxygen (O)-deficient defects that related a peak located at 487 nm, when nitrogen ions substitute for oxygen ions in the ZnO crystal structure, they act as acceptors, increasing the availability of holes and excitons.

Finally, the current–voltage characteristics of the heterojunction p–n junction formed by the N:ZnO thin

film on a silicon substrate at 500 °C were analyzed under both dark and illuminated conditions. An ideal n-factor of 1.23 was determined for the diode, indicating the potential of the N:ZnO sample for application in electronic devices. Overall, the findings underscore the versatility of N:ZnO thin films in optoelectronic applications and the significance of annealing temperature in modulating their structural and optical properties.

Author contributions

S.S.: Conceptualization, Methodology, Supervision, Funding acquisition, Writing-original draft, Project administration. J.S., S.K., M.B., and A.F.: Data curation, Formal analysis, Validation, Writing-original draft. N.T, L.D., K.R, and A.S.S.: Investigation, Resources, Formal analysis, Visualization, Writing-original draft. Ș.Ț.: Writing-review & editing. All authors have read and agreed to the published version of the manuscript.

Funding

Not applicable.

Data availability

All data generated or analyzed during this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interests The authors declare that they have no known competing financial interests or personal relationships that could has appeared to influence the work reported in this paper.

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